

# Where was MH370 at 00:19 UTC?

A reproduction of the DSTG Bayesian aircraft-position pdf (Davey et al. 2016, Fig. 10.3)

*Base estimate: no hypotheses enabled.*

This report recreates the probability density function (pdf) of the aircraft's latitude at the final satellite handshake, 00:19 UTC on 8 March 2014, from the model published by Davey, Gordon, Holland, Rutten and Williams, *Bayesian Methods in the Search for MH370* (Springer, 2016). The aircraft state is filtered from the penultimate radar point at 18:01:49 UTC through all Inmarsat burst timing (BTO) and burst frequency (BFO) measurements, using the published cruise, manoeuvre and measurement models.

With BTO and BFO the recreated pdf has median 37.95°S and mode 37.85°S, with 95% of probability between 33.62°S and 39.50°S. The published curve has median 37.53°S, mode 37.90°S and 95% range 34.90–39.38°S.

The two densities overlap by 76%. This curve is centred 0.4° further south and carries 11% of probability in the published northern shoulder between 34.5°S and 36.5°S (25% in the book). The difference persists at every particle count tried, so it reflects inputs the book does not publish (ACCESS-G weather, the numerical radar prior, autopilot-mode weights) rather than sampling noise.

Relative to the base estimate, the median moves 0.21° north, the 34.5–36.5°S shoulder holds 11% of probability (base 3%), and the two curves overlap by 87%.

The curve pools 4 independent replicates of 7,000,000 particles. Single replicates overlap each other by 77%–85% and their medians span 0.33°. Pooling half of the replicates against the other half gives only 87% overlap, so the pooled curve is not yet converged at this particle count.

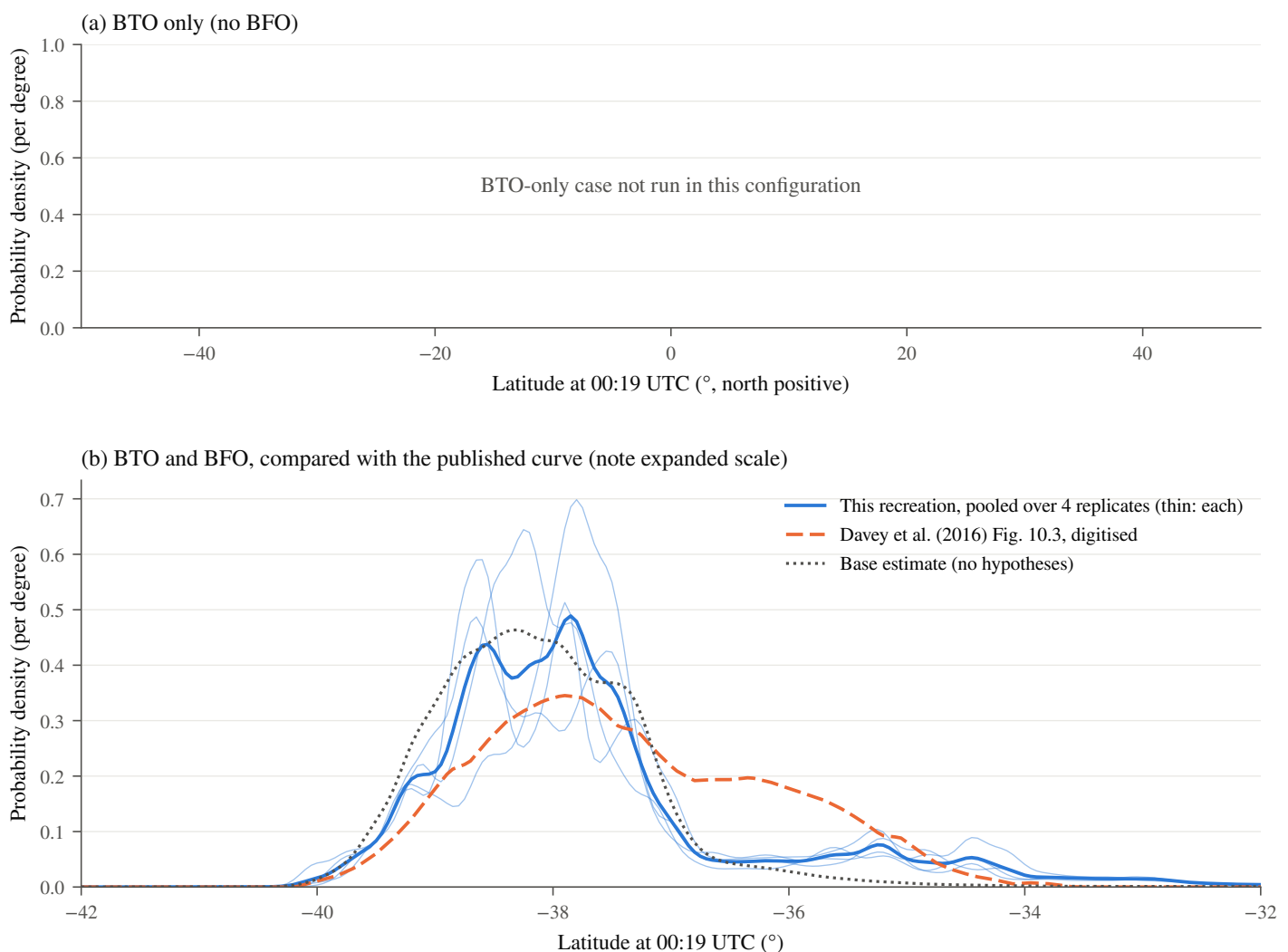


Figure 1. Latitude pdf of the aircraft at 00:19:37 UTC, the form of Davey et al. Fig. 10.3. (a) BTO only. (b) BTO and BFO, with the published curve. Densities are weighted particle histograms smoothed with a 0.1° Gaussian; each particle carries its own altitude. Evidence: BTO at every epoch except the 18:39 and 23:15 C-channel calls; BFO (bottom only) at 18:28, 18:39, 19:41–22:41, 23:15 and 00:11. No fuel, debris, drift, search or end-of-flight evidence is used. Base estimate: no hypotheses enabled.

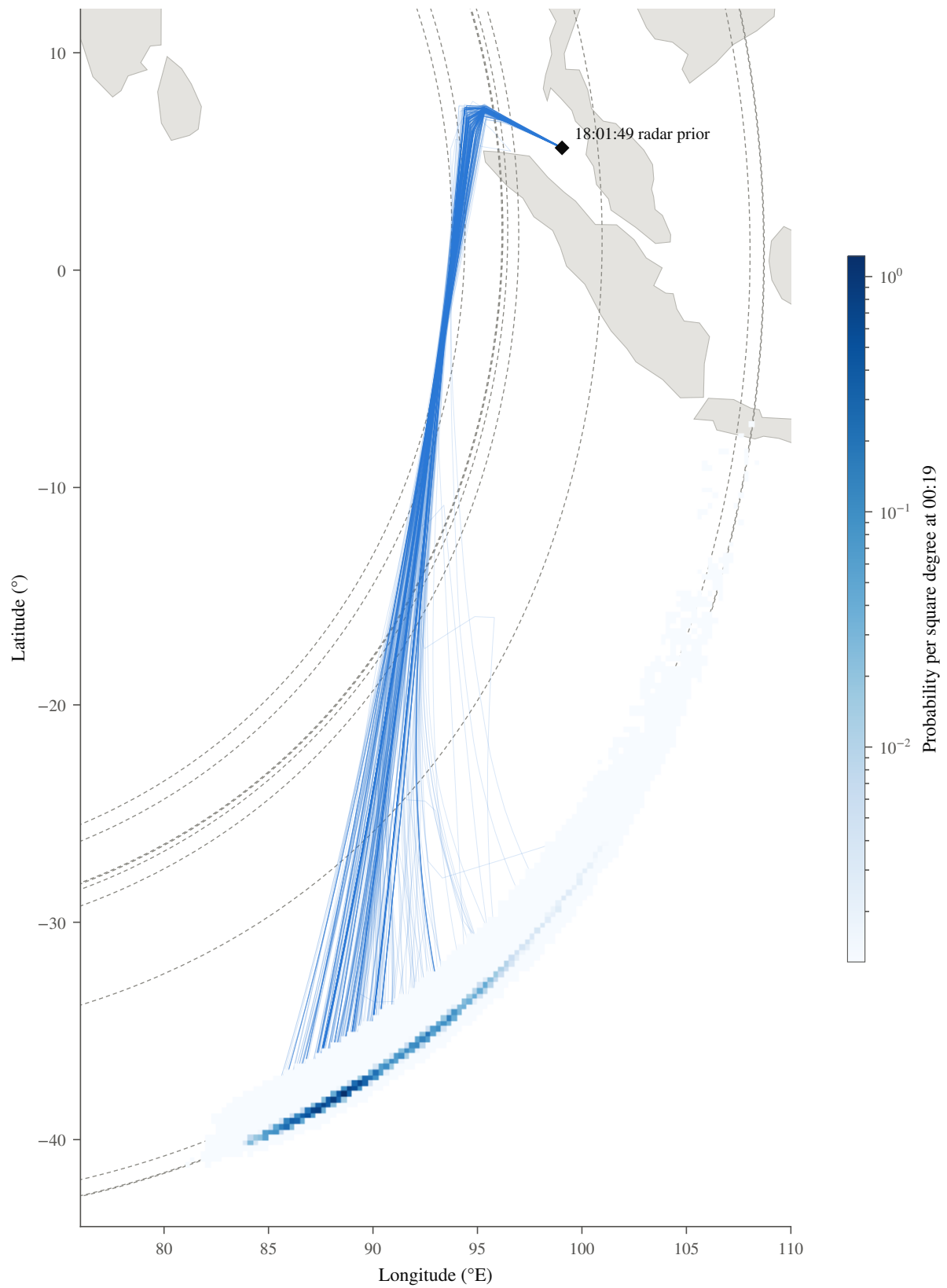
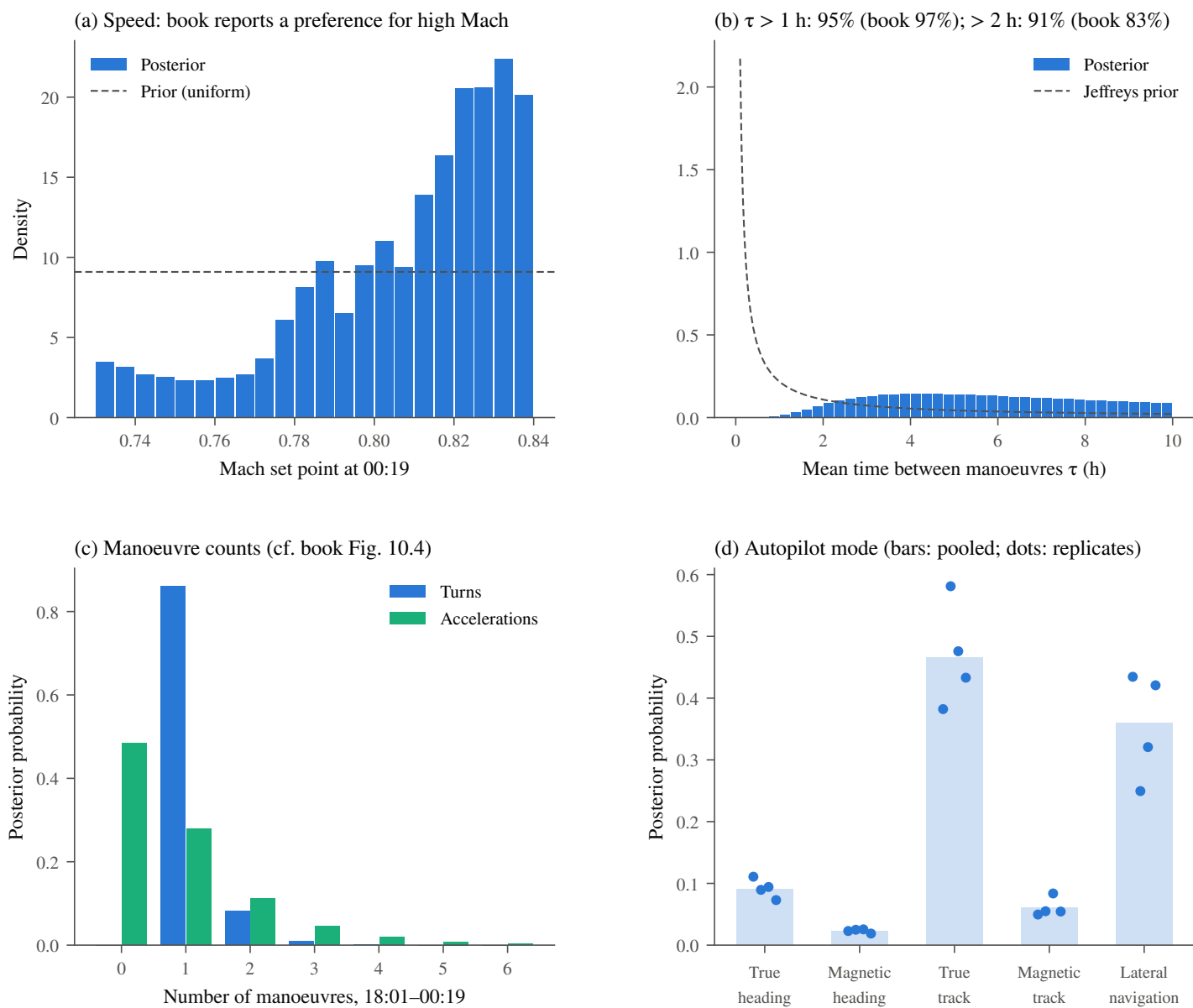


Figure 2. Posterior aircraft position at 00:19 UTC with BTO and BFO, pooled over 4 replicates (blue shading, logarithmic), and 400 routes drawn in proportion to posterior weight from replicate seed 1 (thin lines, 10-minute vertices). Dashed grey curves are the BTO arcs at a reference altitude of 35,000 ft; they are drawn for orientation only. Land: Natural Earth 1:110m.

## Posterior behaviour compared with the book



|                               | This recreation          | Davey et al. (2016)     |
|-------------------------------|--------------------------|-------------------------|
| Median latitude               | 37.95°S                  | 37.53°S                 |
| Mode                          | 37.85°S                  | 37.90°S                 |
| 95% interval                  | 33.62–39.50°S            | 34.90–39.38°S           |
| 95% width                     | 5.88°                    | 4.47°                   |
| Overlap with published pdf    | 76% (replicates 71%–77%) | —                       |
| Median Mach at 00:19          | 0.815                    | upper part of 0.73–0.84 |
| P(north of equator), BTO only | not run                  | northern mode present   |

Figure 3 and Table 1. Posterior summaries with BTO and BFO, pooled over replicates, alongside the statements in Davey et al. (2016), sec. 10.2–10.4. Book values of the latitude pdf come from the digitised Fig. 10.3 (bottom); other book values are quoted from the text.

## Method, convergence and reproduction

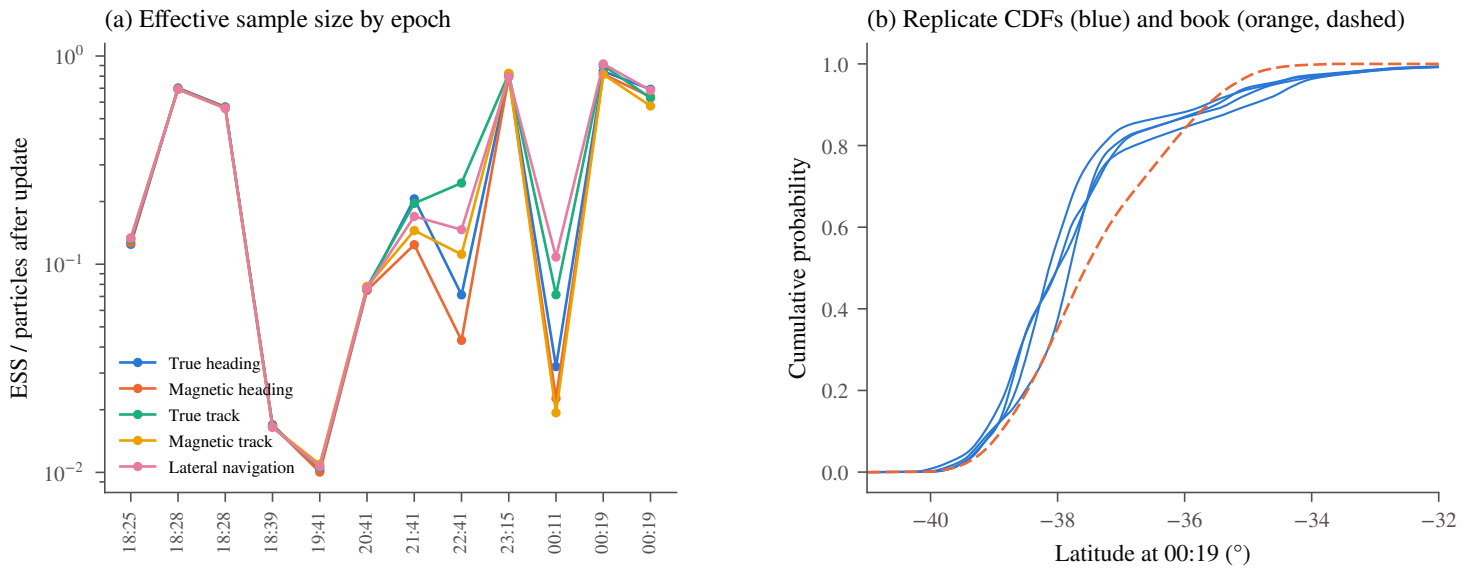


Figure 4. (a) Effective sample size after each measurement update, averaged over replicates, per autopilot-mode filter. The narrowest bottlenecks are the 18:39 C-channel BFO (the southward turn) and the 19:41 BTO arc. (b) Latitude CDFs of the 4 independent replicates; the largest difference between any two is 0.198 (Kolmogorov distance).

Model. The state, dynamics and likelihood follow Davey et al. (2016) chapters 4–8: prior at 18:01:49 UTC with 0.5 NM position and  $1^\circ$  track uncertainty, Mach uniform on 0.73–0.84, altitude on 1,000 ft levels from 25,000 to 43,000 ft; five autopilot modes (true/magnetic heading, true/magnetic track, lateral navigation with a one-off reversion to heading hold); Ornstein–Uhlenbeck fluctuations of Mach, control angle and wind error with the published rates; turns, speed changes and climbs arriving with exponential gaps of common mean  $\tau$  (Jeffreys prior on 0.1–10 h) at  $15^\circ$  bank, 0.1 Mach per minute and 4,000 ft per minute. BTO is Gaussian with the tabulated per-message errors (29, 43 or 63  $\mu$ s); BFO has 7 Hz noise and an unknown constant bias (prior  $150 \pm 25$  Hz) marginalised exactly per particle with a Kalman filter.

Substitutions. The book's ACCESS-G weather is replaced by ERA5 winds and temperatures; magnetic declination is IGRF-14. The prior mean position and track are reconstructed from the book's radar figures because they are not tabulated. Davey et al. used variable-rate branching; this recreation uses fixed-size particle filters with systematic resampling when ESS falls below 50% of the particles. The book warns that conventional resampling collapses static parameters (sec. 10.4). Both are therefore handled exactly: one filter per autopilot mode, combined in proportion to its estimated marginal likelihood, and a Gibbs refresh of  $\tau$  from its path conditional after every resampling step. Without them, a plain bootstrap filter of one million particles retained only about one hundred distinct prior draws, and an earlier recreation in this project found tails and turn counts that changed with the random seed.

Run. Configuration mach-wide; particles per mode 1,000,000, 500,000, 2,500,000, 500,000, 2,500,000 (order as in Figure 3d), 7,000,000 per replicate; BTO+BFO seeds 1, 2, 3, 4; BTO-only seeds ; code revision 0ee1110-dirty; 6 threads; total runtime 31.7 min; peak memory 4,885 MiB. Base estimate: no hypotheses enabled.

Reproduce with one command from the repository root: make report